

Chapter 7—Anatomy and Function of a Gene: Dissection Through Mutation

Fill in the Blank

1. A _____ allele is one whose frequency is greater than 1% in a natural population.
Ans: wild-type
Difficulty: 1
2. _____ are heritable changes in base sequence that can affect phenotype.
Ans: mutations
Difficulty: 1
3. A _____ is a specific protein-encoding segment of DNA composed of a distinctive set of nucleotide pairs.
Ans: gene
Difficulty: 1
4. A mutation that changes a wild-type allele of a gene to a different allele is called a _____ mutation.
Ans: forward
Difficulty: 2
5. A _____ mutation causes a novel allele to be converted back to a wild-type allele.
Ans: reverse
Difficulty: 2
6. An _____ is a type of mutation where a segment of a chromosome is rotated 180°.
Ans: inversion
Difficulty: 1
7. A mound of genetically identical bacteria derived from a single bacterium is called a _____.
Ans: colony
Difficulty: 1
8. Any physical or chemical agent that increases the rate of mutation above the spontaneous rate is a _____.
Ans: mutagen
Difficulty: 1

9. A collection of mutations that *do not* complement each other is known as a _____.
Ans: complementation group
Difficulty: 3
10. A cell that can synthesize all its needed molecules for growth from minimal media is _____.
Ans: prototrophic
Difficulty: 3
11. The end of a polypeptide that contains a free amino group that is not connected to another amino acid is called the _____.
Ans: N-terminus
Difficulty: 1
12. When a protein is exposed to heat, extremes of pH, or molecules such as urea or mercaptoethanol it unfolds and is said to be _____.
Ans: denatured
Difficulty: 2
13. The three-dimensional structure of a protein consisting of two or more polypeptide chains is called _____ structure.
Ans: quaternary
Difficulty: 2
14. A _____ mutation produces either much less of a protein or a protein with very weak but detectable function.
Ans: hypomorphic
Difficulty: 3
15. _____ results when one wild-type allele does not provide enough gene product to produce the wild-type phenotype.
Ans: haploinsufficiency
Difficulty: 2

Multiple Choice

16. The term mutation refers to:
A) only changes in the DNA that result in new phenotypes.
B) only changes in the DNA that result in novel proteins.
C) any change in the DNA of a cell.
D) a heritable change in the DNA of a cell.
E) any change in the cell that changes its survival chances.
Ans: D
Difficulty: 2

17. A heritable change in DNA base sequence is called a:

- A) forward mutation.
- B) reversion.
- C) substitution.
- D) deletion.
- E) mutation.

Ans: E

Difficulty: 1

18. Replacing a thymine nucleotide with a guanine is an example of a:

- A) translocation.
- B) transition.
- C) transversion.
- D) forward mutation.
- E) reversion or reverse mutation.

Ans: C

Difficulty: 1

19. Replacing an adenine nucleotide with a guanine is an example of a:

- A) translocation.
- B) transition.
- C) transversion.
- D) forward mutation.
- E) reversion or reverse mutation.

Ans: B

Difficulty: 1

20. Assume that a wild-type sequence is 5'AGCCTAC3'. Indicate the sequence that might be produced by a transversion.

- A) 5'AGTCTAC3'
- B) 5'AGCCGCCGCCGCCTAC3'
- C) 5'AGCCCAC3'
- D) 5'ATCCTAC3'
- E) 5'AGCCTGC3'

Ans: D

Difficulty: 3

21. A mutation in which parts of two nonhomologous chromosomes change places is called a:

- A) translocation.
- B) transition.
- C) transversion.
- D) insertion.
- E) deletion.

Ans: A

Difficulty: 1

22. Indicate the statement that is most accurate regarding mutations.

- A) Most mutations alter protein structure and phenotype.
- B) Only those mutations that alter protein structure will alter phenotype.
- C) Mutations altering a region that does not code for a protein may alter phenotype.
- D) All mutations that alter protein structure will alter phenotype.
- E) All altered phenotypes result from altered proteins.

Ans: C

Difficulty: 3

23. Assume that the mutation rate for a given gene is 5×10^{-6} mutations per gene per generation. For that gene how many mutations would be expected if 10 million sperm are examined?

- A) none
- B) 5×10^{-6}
- C) 5
- D) 50
- E) 500

Ans: D

Difficulty: 3

24. Which type of mutation is least likely to revert?

- A) deletion
- B) transition
- C) transversion
- D) insertion
- E) all are equally likely

Ans: A

Difficulty: 2

25. Consider the following results. When 50 million sperm were examined for a specific mutation, 100 mutations were found. Indicate the mutation rate for that gene.

- A) 5×10^{-6}
- B) 50×10^{-6}
- C) 2×10^{-6}
- D) 2×10^{-5}
- E) 5×10^{-5}

Ans: C

Difficulty: 2

26. Assume that a researcher set out to duplicate the Luria-Delbruck fluctuation experiment. This researcher planted twenty small flasks with bacteria from the same colony and let them grow overnight. The next morning the researcher noticed that all but one of the flasks had come open and were ruined. Not wishing to redo the experiment the researcher took bacteria samples from the one remaining intact flask and placed them on twenty phage plates. What results would you expect to see when the twenty phage plates are examined, and how would these results compare with those of the original Luria-Delbruck fluctuation experiment?

- A) Identical to the Luria-Delbruck results, namely different numbers of resistant colonies.
- B) Identical to the Luria-Delbruck results, namely identical numbers of resistant colonies.
- C) Not like the Luria-Delbruck results, namely different numbers of resistant colonies.
- D) Not like the Luria-Delbruck results, namely identical numbers of resistant colonies.
- E) Identical to the Luria-Delbruck results, namely no resistant colonies.

Ans: D

Difficulty: 4

27. The results of the Luria-Delbruck fluctuation experiment indicated that:

- A) bacteria are naturally resistant to phage.
- B) a low level of any bacteria population are naturally resistant to phage.
- C) bacteria become resistant to phage by mutation when exposed to phage.
- D) bacteria become resistant to phage by random spontaneous mutation.
- E) the phage mutate to produce large plaques with sharp edges.

Ans: D

Difficulty: 2

28. In the Luria-Delbruck fluctuation experiment, the bacteria + phage plates showed:

- A) all plates had some resistant colonies, some had very many.
- B) some plates had no resistant colonies, a few plates had very many resistant colonies.
- C) all plates had the same number of resistant colonies.
- D) some plates had no resistant colonies, the plates that had resistant colonies all had the same number of resistant colonies.
- E) phage caused mutations to occur in some of the plates but not in others.

Ans: B

Difficulty: 3

29. The hydrolysis of a purine base from the deoxyribose-phosphate backbone is called:

- A) depurination.
- B) deamination.
- C) replica plating.
- D) excision repair.
- E) deletion.

Ans: A

Difficulty: 1

30. Assume that in the organism under study the DNA polymerase has an error rate of 1 mistake in every 10^6 bases copied. However, the overall mutation rate is much lower.

This is most likely because:

- A) the polymerase is more careful in replicating regions where genes exist.
- B) repair mechanisms correct errors made by the polymerase.
- C) not all mutations can be detected easily.
- D) the DNA polymerase has no proofreading function.
- E) mutations do not occur if mutagens are not present.

Ans: B

Difficulty: 2

31. Excision repair corrects DNA by:

- A) removing a double-stranded fragment of damaged DNA.
- B) detecting, removing, and replacing a single stranded fragment of damaged DNA.
- C) excising the incorrect base from a nucleotide.
- D) removing extraneous groups such as methyl or oxygen added by mutagens.
- E) correcting A=T to C=G transitions.

Ans: B

Difficulty: 1

32. Thymine dimers are caused by:

- A) X-rays.
- B) free radicals such as oxygen.
- C) EMS or NSG.
- D) depurination.
- E) UV light.

Ans: E

Difficulty: 1

33. UV light is a mutagen that can cause:

- A) depurination.
- B) deamination.
- C) alkylation.
- D) thymine dimers.
- E) oxidation.

Ans: D

Difficulty: 1

34. The genetic condition xeroderma pigmentosum, which can lead to skin cancer, results from:

- A) inability to correct UV induced dimers.
- B) inability to process phenylamine.
- C) inability to produce functional hemoglobin.
- D) inability to correct transitions.
- E) breaks in the X chromosome.

Ans: A

Difficulty: 2

35. The bacterial repair system that corrects mismatched bases after polymerization is able to discriminate between the old and newly made DNA strands because:

- A) the new strand will contain the incorrect base if a mismatch occurs.
- B) older DNA is more likely to contain errors.
- C) older DNA contains methyl groups at specific sequences.
- D) newer DNA contains methyl groups at specific sequences.
- E) the DNA polymerase is attached to the new strand.

Ans: C

Difficulty: 2

36. The consequence to a bacterial cell of a mutation that inactivated the enzyme that methylates the A of the sequence GATC in newly made DNA would be:

- A) failure to carry out replication.
- B) failure to correct thymine dimers.
- C) failure to distinguish old and new DNA during mismatch repair.
- D) inactivation of certain metabolic genes.
- E) decrease in the mutation rate.

Ans: C

Difficulty: 3

37. Unequal crossing over results in:
- A) an exchange between nonhomologous chromosomes.
 - B) a loss of genetic material.
 - C) a repair of UV-induced damage.
 - D) a production of eggs containing Y chromosomes.
 - E) a creation of deletions and duplications.
- Ans: E
Difficulty: 2
38. The heritable disorder fragile X syndrome, a major cause of mental retardation, is caused by:
- A) production of enzymes that break the phosphate backbone.
 - B) UV light.
 - C) X-rays.
 - D) presence of an extra X chromosome in the sperm or egg.
 - E) duplication of multiple three-nucleotide repeats.
- Ans: E
Difficulty: 2
39. If a man shows the premutation allele for fragile X syndrome, what is the probability that he will pass it on to his son?
- A) 100%
 - B) 75%
 - C) 50%
 - D) 25%
 - E) 0%
- Ans: E
Difficulty: 2
40. The duplication of the triplet sequence CGG resulting in elongation or breakage of the X chromosome is termed:
- A) Barr-eyed.
 - B) Huntington's disease.
 - C) unequal crossing over.
 - D) fragile X syndrome.
 - E) Rhys syndrome.
- Ans: D
Difficulty: 1

41. Genes on the X chromosome of mammals and *Drosophila* are particularly suitable for genetic study because:

- A) males have only one X and most genes behave as haploids.
- B) females have only one X and most genes behave as haploids.
- C) the X chromosome is large and many more genes are located there.
- D) when present as Barr bodies they are exposed for electron microscopic examination.
- E) they behave as diploids in females.

Ans: A

Difficulty: 2

42. If a base analog such as 5-Bromouracil is used as a mutagen, how many generations will be required to mutate the codon for proline (CCC) into the codon for alanine (GCC)?

- A) one generation
- B) two generations
- C) three generations
- D) at least two, but perhaps more due to chance
- E) it will not occur

Ans: E

Difficulty: 3

43. Base analogs differ from other classes of mutagen in that they:

- A) only alter bases.
- B) can only cause transversions.
- C) only work during DNA replication or repair.
- D) can only cause forward mutations, nor reversions.
- E) will not function in bacterial cells.

Ans: C

Difficulty: 2

44. Intercalating agents such as acridine orange function as mutagens to:

- A) promote transitions.
- B) remove amine groups.
- C) attach to purines causing distortions.
- D) add ethyl or methyl groups.
- E) fit between stacked bases and disrupt replication.

Ans: E

Difficulty: 2

45. Alkylating agents such as ethylmethane sulfate (EMS) function as mutagens to:

- A) promote deletions and insertions.
- B) remove amine groups.
- C) add oxygen free radicals to bases.
- D) add ethyl or methyl groups.
- E) fit between stacked bases and disrupt replication.

Ans: D

Difficulty: 2

46. In the Ames test for mutagenicity:

- A) auxotrophic bacteria are converted to prototrophs which survive.
- B) prototrophic bacteria are converted to auxotrophs which survive.
- C) cells are treated with mutagen and only those with no mutations survive.
- D) cells are treated with excess amino acids, killing cells that carry mutations.
- E) rat liver enzymes protect cells from mutation.

Ans: A

Difficulty: 3

47. In the Ames test for mutagenicity, rat liver enzymes are included with the compound under test because:

- A) bacterial cell walls must be treated to permit uptake of the compounds.
- B) rat liver enzymes increase the sensitivity of the bacteria to mutagens.
- C) rat liver enzymes kill mutant cells and allow colonies to form.
- D) rat liver enzymes may modify or break down some compounds.
- E) the mutant strain of bacteria requires rat liver enzymes to digest nutrients for growth.

Ans: D

Difficulty: 2

48. Assume that a new low-calorie sweetener is developed. The structure is novel and is tested with the Ames test for mutagenicity. The following results are obtained:

<u>Sample</u>	<u>Number of his⁺ revertent colonies</u>
distilled water	2
distilled water + rat liver enzymes	3
sweetener	6
sweetener + rat liver enzymes	65

What conclusion is most consistent with this data?

- A) The sweetener is not mutagenic.
- B) Rat liver enzymes are highly mutagenic.
- C) The sweetener is not mutagenic but can be converted into strong mutagens.
- D) The sweetener is mutagenic and can be converted into strong mutagens.
- E) The sweetener and its conversion products are equally mutagenic.

Ans: C

Difficulty: 3

49. The Ames test for mutagenicity is useful to identify potential carcinogens because:
- A) since bacteria do not get cancer they can survive lethal carcinogens.
 - B) mutagens that affect bacterial DNA are likely to cause human mutation.
 - C) bacteria thrive on substances that could cause cancer in humans.
 - D) the same genes that cause cancer in humans can be mutated in bacteria.
 - E) liver enzymes alter the bacteria so they will behave like mammal cells.

Ans: B

Difficulty: 2

50. The size of the human genome in base-pairs is about:

- A) 210 million
- B) 100,000
- C) 2.75×10^6
- D) 2.75×10^9
- E) 2.75×10^{10}

Ans: D

Difficulty: 1

51. A complementation group is:

- A) a group of mutations that produce the same phenotype.
- B) a group of mutations that are in the same gene and complement each other.
- C) a group of mutations that are in the same gene and do not complement each other.
- D) a group of mutations in two different genes that complement each other.
- E) a group of mutations in two different genes that do not complement each other.

Ans: C

Difficulty: 3

52. Choose the statement that is most correct regarding the rII⁻ strain of T4 that Benzer studied.

- A) Produces smaller plaques than wild type.
- B) Produces smaller plaques, grows in *E. coli* K(λ), not in *E. coli* B.
- C) Produces larger plaques, grows in *E. coli* K(λ), not in *E. coli* B.
- D) Produces larger plaques, grows in *E. coli* B, not in *E. coli* K(λ).
- E) Produces larger plaques, grows in both *E. coli* K(λ) and in *E. coli* B.

Ans: D

Difficulty: 3

53. A plaque is:

- A) a colony of bacteria growing on a plate.
- B) a colony of bacteria that contain phage within them.
- C) a region on a plate where living bacteria survive phage infection.
- D) an area on a plate containing live phage-resistant bacteria.
- E) an area on a plate containing phage and dead or destroyed bacteria.

Ans: E

Difficulty: 1

54. Shown below are the results of a series of coinfections using T4 rII- strains similar to those employed by Benzer. Each strain contains a different deletion mutation. Ability to produce wild-type progeny phage is indicated by (+), (o) indicates no wild-type progeny.

	A	B	C	D	E
A	o	+	o	+	+
B	+	o	+	o	o
C	o	+	o	+	+
D	o	o	+	o	+
E	+	o	+	+	o

- A) CADBE
- B) ACBDE
- C) ABCDE
- D) BEDCA
- E) CEADB

Ans: A

Difficulty: 3

55. Shown below are the results of a series of coinfections using T4 rII- strains similar to those employed by Benzer. Each strain contains a different deletion mutation. Ability to produce wild-type progeny phage is indicated by (+), (o) indicates no wild-type progeny.

	A	B	C	D	E
A	o	o	+	o	+
B	o	o	+	o	+
C	+	+	o	o	o
D	o	+	o	o	+
E	+	+	o	+	o

Indicate the order that is most consistent with these data.

- A) CADBE
- B) ACBDE
- C) BADCE
- D) BEDCA
- E) CEADB

Ans: C

Difficulty: 3

56. Choose the statement that best distinguishes a complementation test and a recombination analysis when examining mutations in phage.
- A) Both tests require two different mutations.
 - B) Recombination can only occur between two genes.
 - C) Complementation results can be seen immediately, recombination requires a second infection.
 - D) Recombination results can be seen immediately, complementation requires a second infection.
 - E) Recombination can distinguish one gene with two alleles from two different genes.

Ans: C

Difficulty: 3

57. Shown below are the deletion maps of a series of rII- mutations. The deleted region is indicated as (.....) and the intact region as _____.

```

1  _____(.....)_____
2  _____(.....)_____
3  (.....)_____
4  _____(.....)_____
5  _____(.....)_____

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A series of point mutations A-E is used in a coinfection experiment. Shown below are the results of those coinfections. Ability to produce wild-type progeny phage is indicated by (+), (o) indicates no wild-type progeny.

	1	2	3	4	5
A	+	+	o	+	+
B	+	+	+	+	+
C	+	+	+	o	+
D	+	+	+	+	o
E	+	o	+	+	+

Indicate the order that is most consistent with these data.

- A) CADBE
- B) ACBDE
- C) BADCE
- D) ABDEC
- E) CEADB

Ans: D

Difficulty: 4

58. Shown below are the deletion maps of a series of rII- mutations. The deleted region is indicated as (.....) and the intact region as _____.

1 _____(.....)_____

2 _____(.....)_____

3 (.....)_____

4 _____(.....)

5 _____(.....)_____

A series of point mutations A-E is used in a coinfection experiment. Shown below are the results of those coinfections. Ability to produce wild-type progeny phage is indicated by (+), (o) indicates no wild type progeny.

	1	2	3	4	5
A	+	o	+	+	+
B	o	+	+	+	+
C	+	+	+	o	+
D	+	+	o	+	+
E	+	+	o	+	o

Indicate the order that is most consistent with these data.

- A) CADBE
- B) DEBAC
- C) BADCE
- D) ABDEC
- E) CEADB

Ans: B

Difficulty: 3

59. Shown below are the deletion maps of a series of rII⁻ mutations. The deleted region is indicated as (.....) and the intact region as _____. Note that strain 5 carries two different deletions.

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1  _____(.....)_____
2  _____(.....)_____
3  (.....)_____
4  _____(.....)_____
5  _____(.....)_____

```

A series of point mutations A-E is used in a coinfection experiment. Shown below are the results of those coinfections. Ability to produce wild-type progeny phage is indicated by (+), (o) indicates no wild type progeny.

	1	2	3	4	5
A	+	+	o	+	o
B	+	o	+	+	+
C	+	+	o	+	+
D	o	+	+	+	+
E	+	+	+	o	o

Indicate the order that is most consistent with these data.

- A) CADBE
- B) DEBAC
- C) BADCE
- D) ABDEC
- E) CEADB

Ans: A

Difficulty: 4

60. Indicate the correct order for one round of infection by bacteriophage T4.

1. Lysis of host cell.
2. Phage proteins and DNA synthesized, host DNA degraded.
3. Assembly of phage within host cell.
4. Phage body enters host cell.
5. Phage injects DNA into host cell.

- A) 4, 2, 3, 1
- B) 1, 2, 3, 4, 5
- C) 5, 1, 2, 3,
- D) 5, 2, 3, 1
- E) 4, 5, 3, 1

Ans: D

Difficulty: 2

61. How many progeny phage are released when a single *E. coli* cell is lysed by phage T4?
- A) between 1 and 10
 - B) between 10 and 100
 - C) between 100 and 1,000
 - D) about 10,000
 - E) about 100,000

Ans: C

Difficulty: 1

62. Indicate which of the following is least important in doing a complementation test with coinfection of phage T4.
- A) Ensuring that sufficient phage of both strains are present.
 - B) Recovering phage from the plaques after growth and lysis.
 - C) Counting the plaques that are produced on *E. coli* K(λ).
 - D) Control using both mutations in cis configuration and a wild type.
 - E) All the above steps are essential for the experiment.

Ans: C

Difficulty: 3

63. Assume that a researcher is studying coat color in voles. Three strains of white vole have been isolated: milky, blanc, and weiss. White is a recessive trait in each strain. Homozygous white voles are obtained for each strain. Consider the following crosses:
- milky $\times$ blanc = all white progeny
 - milky $\times$ weiss = all brown (wild-type vole color)
 - blanc $\times$ weiss = all brown (wild-type vole color)

The conclusion most consistent with these results is:

- A) all three strains have mutations in the same gene.
- B) all three strains have mutations in different genes.
- C) milky and blanc have mutations on the same gene, weiss has a mutation in a different gene.
- D) milky and weiss have mutations on the same gene, blanc has a mutation in a different gene.
- E) weiss and blanc have mutations on the same gene, milky has a mutation in a different gene.

Ans: C

Difficulty: 3

64. Assume a researcher is studying the rII locus of phage T4. Three rII⁻ strains are obtained: A, B, and C. When coinfections are performed in *E. coli* strain K(λ) the following results are obtained:

A × B = plaques form

A × C = plaques form

B × C = no plaques form

The conclusion most consistent with these data is:

- A) A, B, and C carry mutations in three different genes.
- B) A and B carry mutations in the same gene, C is on a different gene.
- C) A and C carry mutations in the same gene, B is on a different gene.
- D) B and C carry mutations in the same gene, A is on a different gene.
- E) A, B, and C carry mutations in the same gene.

Ans: D

Difficulty: 3

65. Assume a researcher is studying the rII locus of phage T4. Four rII⁻ strains are obtained: A, B, C and D. When coinfections are performed in *E. coli* strain K(λ) the following results are obtained:

A × B = lysis

A × C = lysis

B × C = no lysis

B × D = no lysis

C × D = no lysis

In a second experiment, coinfections are performed in *E. coli* strain B. When progeny phage are examined for their ability to form plaques in *E. coli* strain K(λ), the following results are obtained:

A × B = plaques

B × C = plaques

C × D = plaques

B × D = no plaques

The conclusion most consistent with these data is:

- A) A carries a mutation in one gene, B, C and D are on a different gene, C and D both carry the same mutation.
- B) A and B carry mutations in the same gene, C and D are on a different gene.
- C) A carries a mutation in one gene, B, C and D are on a different gene, B and C both carry the same mutation.
- D) A carries a mutation in one gene, B, C and D are on a different gene, B and D both carry the same mutation.
- E) A, B, C and D carry mutations in the same gene.

Ans: D

Difficulty: 4

66. Choose the statement that is most accurate concerning biochemical pathways.
- A) All enzymes in the pathway catalyze the same reaction.
 - B) If an enzyme in a pathway is inactive, adding excessive amounts of its substrate will restore the normal phenotype.
 - C) If an enzyme in a pathway is inactive, adding excessive amounts of its product will restore the normal phenotype.
 - D) If the enzyme that catalyzes the final step in a pathway is inactive all the other enzymes will be inactivated as well.
 - E) If the first enzyme in a pathway is inactivated, adding the final product will not restore the normal phenotype.

Ans: C

Difficulty: 2

67. Assume 7 different strains of fly have been isolated, each shows a recessive white eye trait. Crosses are performed as follows; (w) indicates white-eyed progeny, (R) indicates wild-type red eyes.

	A	B	C	D	E	F	G
A		w	w	R	R	w	R
B			w	R	R	w	R
C				w	w	R	R
D					w	R	R
E						w	R
F							w
G							

Based on these crosses, how many different genes are present?

- A) only one gene with several different alleles
- B) 2
- C) 3
- D) 4
- E) 7

Ans: C

Difficulty: 3

68. Assume 8 different strains of fly have been isolated, each shows a recessive white eye trait. Crosses are performed as follows; (w) indicates white-eyed progeny, (R) indicates wild-type red eyes.

	A	B	C	D	E	F	G	H
A		w		R	w	R	R	w
B			w	R	w	R	R	w
C				w	R	R	R	R
D					R	R	R	R
E						R	R	w
F							w	R
G								R
H								

Based on these crosses, how many different genes are present and what strains have mutations in the same gene as does strain A?

- A) 2, B, E, and H
- B) 3, B and C
- C) 3, B, C, and H
- D) 3, B, E, and H
- E) 4, B and H

Ans: D

Difficulty: 4

69. In the human genetic disorder alkaptonuria, urine turns black because of the presence of homogentisic acid in individuals with the trait. This is due to:
- A) the presence of large amounts of homogentisic acid in the diet.
 - B) failure of individuals with alkaptonuria to manufacture enzymes involved in the synthesis of homogentisic acid.
 - C) failure of wild-type individuals to manufacture enzymes involved in the synthesis of homogentisic acid.
 - D) failure of the kidneys to remove homogentisic acid from the urine.
 - E) failure of individuals with alkaptonuria to manufacture enzymes involved in the breakdown of homogentisic acid.

Ans: E

Difficulty: 2

70. Consider the pathway for the synthesis of the amino acid arginine in *Neurospora*:

ARG-E ARG-F ARG-H
 ornithine → citrulline → argininosuccinate → arginine

Mutant strains of *Neurospora* are grown in minimal media supplements as follows. Each mutant strain carries only a single mutation. Growth is shown by (+), no growth is shown by (o).

mutant strain	<u>Supplements</u>				
	nothing	ornithine	citrulline	arginino-succinate	arginine
a	o	o	o	+	+
b	o	o	+	+	+
c	o	o	o	o	+

Indicate the correct strain / defective gene pairing.

- A) strain a / ARG-F
- B) strain a / ARG-H
- C) strain b / ARG-F
- D) strain c / ARG-E
- E) strain c / ARG-F

Ans: A

Difficulty: 3

71. Consider the pathway for the synthesis of the amino acid arginine in *Neurospora*:

ARG-E ARG-F ARG-H
 ornithine → citrulline → argininosuccinate → arginine

Mutant strains of *Neurospora* are grown in minimal media supplements as follows. Strains may carry more than one mutation. Growth is shown by (+) no growth is shown by (o).

mutant strain	<u>Supplements</u>				
	nothing	ornithine	citrulline	arginino-succinate	arginine
a	o	o	o	o	+

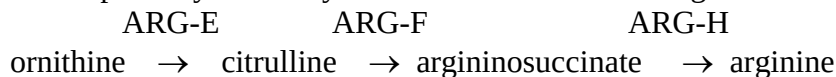
Indicate the most accurate statement regarding strain A.

- A) There is a mutation in ARG-H, if citrulline accumulates, ARG-F is also defective.
- B) There is a mutation in ARG-H, if ornithine accumulates, ARG-F is also defective.
- C) There is a mutation in ARG-H, if argininosuccinate accumulates, ARG-F is also defective.
- D) There is a mutation in ARG-H, if citrulline accumulates, ARG-E is also defective.
- E) There is a mutation in ARG-E, if citrulline accumulates, ARG-F is also defective.

Ans: A

Difficulty: 4

72. Consider the pathway for the synthesis of the amino acid arginine in *Neurospora*:



Mutant strains of *Neurospora* are grown in minimal media supplements as follows. The strains may carry more than one mutation. Growth is shown by (+) no growth is shown by (o).

mutant strain	<u>Supplements</u>				
	nothing	ornithine	citrulline	arginino-succinate	arginine
a	o	o	o	+	+
b	o	o	o	+	+

Strain (a) accumulates citrulline, strain b does not. Indicate the statement that is most correct regarding these two strains.

- A) Strain a has a mutation in ARG-E only.
- B) Strain b has only one mutation.
- C) Strain a has mutations in ARG-F and ARG-H.
- D) Strain a has mutations in ARG-E, ARG-F and ARG-H.
- E) Strain a has a mutation in ARG-H only.

Ans: E

Difficulty: 4

73. Indicate the false statement regarding amino acids.

- A) Every amino acid contains a carboxyl group.
- B) The side chain or R group differs for each amino acid.
- C) Amino acids are joined together by peptide bonds.
- D) The end of the polypeptide termed the N terminus contains a free amino group.
- E) All the above statements are correct.

Ans: E

Difficulty: 1

74. Indicate the false statement regarding amino acids.

- A) Several amino acids linked together are termed an oligopeptide.
- B) Amino acids are linked by peptide bonds that join two amino groups together.
- C) The C terminus of a polypeptide chain contains a free carboxylic acid group.
- D) Two amino acids joined together is termed a dipeptide.
- E) All the above statements are correct.

Ans: B

Difficulty: 2

75. Choose the condition below that does not involve a defect in an enzyme pathway.

- A) alkaptonuria
- B) albinism
- C) sickle-cell anemia
- D) phenylketonuria (PKU)
- E) all of the above involve a defect in an enzyme pathway

Ans: C

Difficulty: 2

76. Choose the interaction listed below that is not involved in maintaining tertiary structure in protein molecules.

- A) covalent bond
- B) hydrogen bond
- C) hydrophobic/hydrophilic interactions
- D) ionic interactions
- E) all of the above may be involved in maintaining protein tertiary structure

Ans: E

Difficulty: 1

77. The condition sickle-cell anemia is due to:

- A) the insertion of an amino acid.
- B) the deletion of an amino acid.
- C) substitution of an amino acid.
- D) failure to synthesize a hemoglobin molecule.
- E) unequal recombination resulting in the deletion of the β -chain hemoglobin gene.

Ans: C

Difficulty: 2

78. Choose the statement below that is not true regarding sickle-cell anemia.

- A) Individuals who are heterozygous for the sickle cell allele can not make hemoglobin.
- B) The sickle-cell hemoglobin molecule contains an amino acid substitution.
- C) The hemoglobin molecules of an individual with sickle cell anemia clump together.
- D) The red blood cells of an individual with sickle cell anemia distort and elongate.
- E) All of the above are true regarding sickle cell anemia.

Ans: A

Difficulty: 2

79. Though sickle-cell anemia is frequently lethal for individuals who are homozygous for the sickle cell allele, natural selection seems to have maintained that allele in certain geographic locations. A likely explanation for this observation is:
- A) the forward mutation rate to sickle-cell is much higher in those regions.
 - B) individuals with sickle-cell anemia live longer and have more children.
 - C) reversion from sickle-cell to wild type is prevented in some populations.
 - D) individuals who are heterozygous for the sickle-cell allele are protected from malaria.
 - E) only certain populations have been tested for the presence of the sickle cell allele.

Ans: D

Difficulty: 2

80. The structure of a polypeptide that is characterized by a three dimensional shape with a characteristic geometry at local regions maintained by hydrogen bonds is:

- A) primary structure.
- B) secondary structure.
- C) tertiary structure.
- D) quaternary structure.
- E) both tertiary and quaternary structure.

Ans: B

Difficulty: 1

81. The structure of a protein that involves the interaction between two distinct polypeptide chains is:

- A) primary structure.
- B) secondary structure.
- C) tertiary structure.
- D) quaternary structure.
- E) both primary and secondary structure.

Ans: B

Difficulty: 1

82. Assume that a certain strain of bacteria carries a mutation that causes it to die at high temperature (37°C), but grows normally at cooler temperatures. This mutation is termed:

- A) recessive.
- B) deletion.
- C) biochemical pathway mutant.
- D) conditional lethal that grows under restrictive conditions.
- E) conditional lethal that dies under restrictive conditions.

Ans: E

Difficulty: 2

83. Mutations that abolish the function encoded by the wild-type allele are known as:

- A) null mutations.
- B) hypomorphic mutations.
- C) hypermorphic mutations.
- D) conditional mutations.
- E) neomorphic mutations.

Ans: A

Difficulty: 1

84. A neomorphic mutation results in an allele that:

- A) produces no gene product.
- B) produces a nonfunctional gene product.
- C) produces novel proteins or cause inappropriate expression resulting in a new phenotype.
- D) produces proteins that aggregate with wild-type subunits, inactivating them.
- E) produces an altered protein that results in a wild-type phenotype.

Ans: C

Difficulty: 2

85. Assume that a transition mutation results in an amino acid substitution in the resulting polypeptide. What level of protein structure might be affected as a result?

- A) primary structure
- B) secondary structure
- C) tertiary structure
- D) quaternary structure
- E) all levels might be affected by a single amino acid substitution

Ans: E

Difficulty: 2

86. The photoreceptor protein rhodopsin:

- A) is found in cone cells and is sensitive to weak light at many wavelengths.
- B) is found in rod cells and is sensitive to weak light at many wavelengths.
- C) is found in cone cells and is responsible for blue and green color vision.
- D) is found in rod cells and is responsible for blue and green color vision.
- E) is missing in individuals who exhibit red-green colorblindness.

Ans: B

Difficulty: 2

87. Examination of the rhodopsin gene family provides evidence for gene evolution by:

- A) duplication and divergence.
- B) accumulation of random mutations.
- C) convergent evolution.
- D) spontaneous generation.
- E) drift.

Ans: A

Difficulty: 1

88. Red-green color blindness is more common in males than females because:

- A) the red pigment gene is on the X chromosome, the green is on an autosome.
- B) the green pigment gene is on the X chromosome, the red is on an autosome.
- C) rhodopsin gene is on the X chromosome.
- D) both the red and the green pigment genes are on the X chromosome.
- E) both the red and the green pigment genes are on an autosome.

Ans: D

Difficulty: 2

89. Consider the gene for color in a particular flower. Three alleles exist: dark blue, white, and sky. Flowers homozygous for each allele produce the characteristic color. In heterozygotes, incomplete dominance occurs and intermediate colors are seen in a natural population. How many different color phenotypes are possible with these three alleles?

- A) 9
- B) 2
- C) 4
- D) 6
- E) 8

Ans: D

Difficulty: 2

90. The appearance of a novel phenotype resulting from the substitution of a single base pair might be due to:

- A) change in the amino acid sequence only.
- B) change in the amount of protein expressed.
- C) alteration in a gene that codes for a nontranslated RNA.
- D) change in the developmental time or location at which a gene is expressed.
- E) all of the above are possible consequences of a single base pair substitution.

Ans: E

Difficulty: 2

91. Assume that a series of compounds has been discovered in *Neurospora*. Compounds A-F appear to be members of an enzyme pathway. Several mutations have been identified and strains 1-4 each contain a single mutation. Shown below are 5 possible pathways. Choose the pathway that best fits the data presented. [growth in minimal media with supplements is shown by (+), no growth is shown by (o)]

	<u>media supplement</u>					
strain	A	B	C	D	E	F
1	o	o	o	+	o	o
2	o	o	o	o	o	+
3	o	o	o	o	+	+
4	o	o	+	o	o	o

- A) $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F$
 B) $A \rightarrow B \rightarrow C \rightarrow F \rightarrow E \rightarrow D$
 C) $A \rightarrow B \rightarrow C \rightarrow D$

↓

$E \rightarrow F$

- D) $A \rightarrow B \rightarrow C \rightarrow D$

↓

$E \rightarrow F$

- E) $A \rightarrow B \rightarrow C \rightarrow D$

↓

$E \rightarrow F$

Ans: E

Difficulty: 4

92. Assume that a series of compounds has been discovered in *Neurospora*. Compounds A-F appear to be members of an enzyme pathway. Several mutations have been identified and strains 1-4 each contains a single mutation. Shown below are 5 possible pathways. Choose the pathway that best fits the data presented. [growth in minimal media with supplements is shown by (+), no growth is shown by (o)]

	<u>media supplement</u>					
strain	A	B	C	D	E	F
1	o	o	o	+	+	+
2	o	o	o	o	+	+
3	o	o	o	o	+	o
4	o	o	+	+	+	+

- A) $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F$
 B) $A \rightarrow B \rightarrow C \rightarrow F \rightarrow D \rightarrow E$
 C) $F \rightarrow B \rightarrow C \rightarrow D \rightarrow A \rightarrow E$
 D) $A \rightarrow B \rightarrow C \rightarrow D \rightarrow F \rightarrow E$
 E) $A \rightarrow B \rightarrow F \rightarrow E \rightarrow C \rightarrow D$

Ans: D

Difficulty: 3

93. The term fecundity refers to:

- A) ability to live a long life.
- B) ability to survive in several different conditions.
- C) ability to transcribe DNA.
- D) ability to produce offspring.
- E) ability to metabolize several different sugar molecules.

Ans: D

Difficulty: 2

94. Assume that for a given gene a mutation creates an allele that functions as a dominant negative. The gene codes for a protein that forms a trimer within the cell. If at least one of the subunits has the mutant structure the entire protein is inactivated. For a heterozygous individual, what percent of the trimers present in the cell will be inactive?

- A) 100%
- B) 25%
- C) 50%
- D) 6.25%
- E) 12.5%

Ans: E

Difficulty: 3

95. A neomorphic dominant mutation in the *Antennapedia* (*Antp*) gene of *Drosophila* causes:

- A) kinks to form in their tails.
- B) shortened tails.
- C) total loss of color vision.
- D) failure to assemble microtubules during mitosis.
- E) growth of leg from the head region.

Ans: E

Difficulty: 2

Matching

Match the following descriptions with the terms that best fit

- a) auxotrophic
- b) prototrophic
- c) null mutation
- d) inversion
- e) transposable element
- f) intercalators
- g) complementation
- h) dominant negative mutation
- i) cistron
- j) reverse mutation

96. _____ mutant allele is mutated to wild-type
Ans: j
Difficulty: 1
97. _____ capable of growth on minimal media
Ans: b
Difficulty: 2
98. _____ complementation group identified by *cis-trans* test
Ans: i
Difficulty: 2
99. _____ grows on minimal media only if supplemented
Ans: a
Difficulty: 1
100. _____ flat planar molecules that sandwich between DNA bases
Ans: f
Difficulty: 1
101. _____ 180° rotation of a segment of a DNA molecule
Ans: d
Difficulty: 1
102. _____ when a mutant subunit of a multimer blocks the activity of the subunits produced by normal alleles
Ans: h
Difficulty: 2
103. _____ mutation that abolishes the function of a protein
Ans: c
Difficulty: 1
104. _____ all DNA segments that move about in a genome, regardless of mechanism
Ans: e
Difficulty: 2
105. _____ when alleles on each of two homologs make up for a defect in the other chromosome, generating enough of both gene products to yield a wild-type phenotype.
Ans: g
Difficulty: 2

True or False

106. Methyl-directed mismatch repair corrects mistakes in replication by methylating the new daughter strand and thus being able to distinguish it from the parental DNA strand.
Ans: False
Difficulty: 2
107. Base analogs similar to the normal nitrogenous bases of DNA can be incorporated during replication often causing substitutions in the next round of replication.
Ans: True
Difficulty: 2
108. The Food and Drug Administration assesses whether an agent is mutagenic by first screening it in an Ames Test and then subsequently in rodents.
Ans: True
Difficulty: 1
109. A cistron is any complementation group identified by the cis-trans test and is synonymous with a gene.
Ans: True
Difficulty: 2
110. A “hot spot” is a region of DNA resistant to mutations.
Ans: False
Difficulty: 1
111. A mutation in a single gene that codes for a protein in a biochemical pathway often leads to auxotrophy for the end product of that pathway.
Ans: False
Difficulty: 1
112. A missense mutation is one that creates a premature stop codon.
Ans: False
Difficulty: 1
113. The secondary structure of a protein is its overall three-dimensional structure.
Ans: False
Difficulty: 1
114. Only multimeric proteins have quaternary structure.
Ans: True
Difficulty: 1

115. Ectopic expression is when a protein is produced outside of its normal place or time.
Ans: True
Difficulty: 2

Short Answer

116. Distinguish between neomorphic, hypomorphic, and hypermorphic mutations.
Ans: A hypomorphic mutation produces either much less of a protein or a protein with very weak but detectable function while a hypermorphic mutation does the opposite; it gives rise to more wild-type protein or the same amount of a more efficient protein. Neomorphic mutations are ones that simply result in a new phenotype.
Difficulty: 3
117. What is a null mutation?
Ans: Null mutations abolish the function of a protein.
Difficulty: 2
118. Geneticists have detected mutations affecting vision that result in blue cone monochromacy. What is this and how does it occur?
Ans: Blue cone monochromacy is an X-linked genetic disorder that affects the function of red and green cones, which results in red/green colorblindness. Seven different deletion mutations have been identified that result in blue cone monochromacy.
Difficulty: 3
119. How is DNA altered by hydrolysis, radiation, UV light, and oxidation respectively?
Ans: DNA hydrolysis of A or G bases results in depurination and the DNA strand has a continuous sugar backbone but an unspecified base where the depurination occurred. X-irradiation breaks the sugar backbone while UV light induces thymidine dimerization. Oxygen free radicals oxidize bases into analogs that do not hydrogen bond properly in the DNA double strand. During replication, mismatch pairing ends up creating a base change resulting in mutation.
Difficulty: 4
120. What mutation occurs in the genetic disease *xeroderma pigmentosa* and what are the consequences?
Ans: Individuals with the disease *xeroderma pigmentosa* have a mutation in excision repair, which results in a lack of enzymatic ability to recognize and repair thymidine dimers.
Difficulty: 2

Experimental Design and Interpretation of Data

121. What two hypotheses are tested in the Luria-Delbruck fluctuation test?

Ans: Hypothesis 1: Resistance is a physiological response

Hypothesis 2: Resistance arises from random mutation

Difficulty: 2

122. What technique would you use to test the hypothesis that multiple drug resistant bacterium exist in a heterogenous population?

Ans: Replica plating on each of the drugs to be screened will indicate the drug resistance of individual bacteria within the population.

Difficulty: 2

123. Chemical X has just been screened using the Ames test. A total of 5000 bacteria were tested against 0.001 mM, 1 mM, 0.1M and 1M concentrations for which 4, 1, 0, and 200 colonies grew respectively. Control plate of minimal media supplemented with histidine had 5000 colonies while minimal media alone had only two. Interpret these data.

Ans: The control plate supplemented with histidine has 5000 colonies which indicates the total number of bacteria present in the sample. The control plate with no histidine has 2 colonies indicating that the natural rate of *his*⁻ reversion is 2/5000. Only the high concentration of 1M chemical X caused a *his*⁻ reversion at a rate significantly higher than control indicating the chemical X is a mutagen only at high levels.

Difficulty: 3

124. The local pet store received several shipments of albino ferrets. You choose two males and two females as pets; one breeding pair from the same litter, one from two different litters. When your ferret's litters are born, one has normally pigmented offspring. State which offspring are albino and which are pigmented and explain why.

Ans: The breeding pair from the same litter would have albino offspring (they would carry a mutation in the same gene) while the breeding pair with the unrelated male and female could have pigmented offspring if each had a mutation in different genes involved in pigmentation. The two unrelated albino ferret's mutations complemented each other's genetic deficiency leading to pigmented offspring.

Difficulty: 3